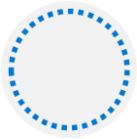


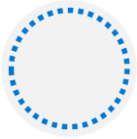
Training Course

Module 06: Azure Storage

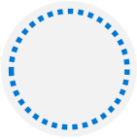




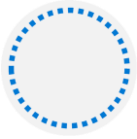
Lesson 01: Configure Storage Accounts



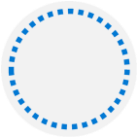
Lesson 02: Configure Blob Storage



Lesson 03: Configure Storage Security

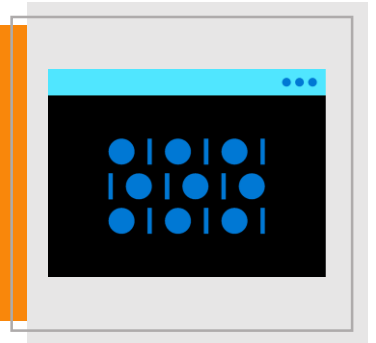


Lesson 04: Configure Azure Files and File Sync

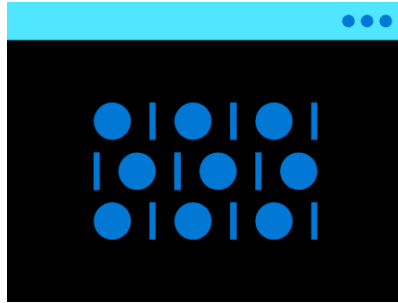


Lesson 05: Configure Storage with Tools

Lesson 01: Configure Storage Accounts



Implement Azure Storage



A service that you can use to store files, messages, tables, and other types of information

Durable, secure,
scalable, managed,
accessible

Storage for virtual
machines, unstructured
data and structured data

Two tiers: Premium
and Standard

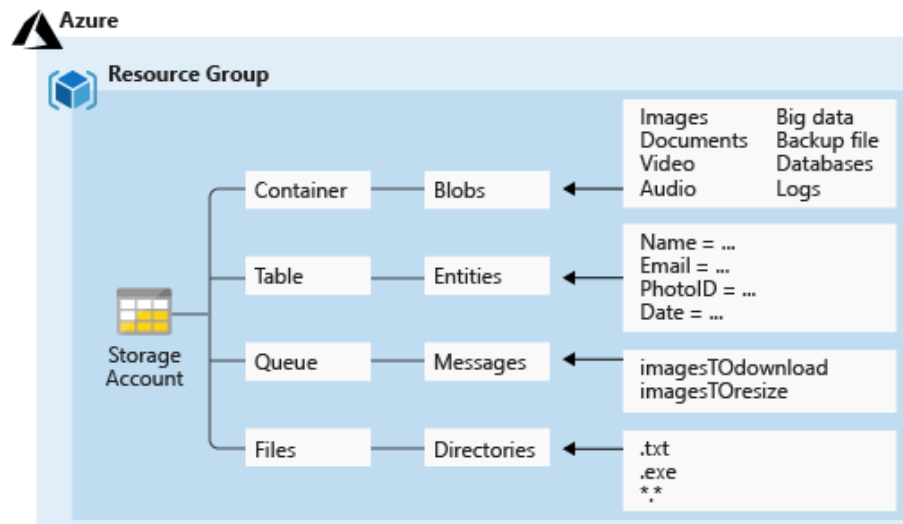
Explore Azure Storage Services

Azure Containers: A massively scalable object store for text and binary data

Azure Tables: Ideal for storing structured, non-relational data

Azure Queues: A messaging store for reliable messaging between application components

Azure Files: Managed file shares for cloud or on-premises deployment



Determine Storage Account Kinds

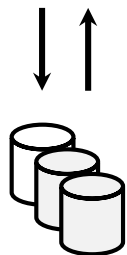
Storage Account	Recommended usage
Standard general-purpose v2	Most scenarios including Blob, File, Queue, Table, and Data Lake Storage.
Premium block blobs	Block blob scenarios with high transactions rates, or scenarios that use smaller objects or require consistently low storage latency.
Premium file shares	Enterprise or high-performance file share applications.
Premium page blobs	Premium high-performance page blob scenarios.



All storage accounts are encrypted using Storage Service Encryption (SSE) for data at rest

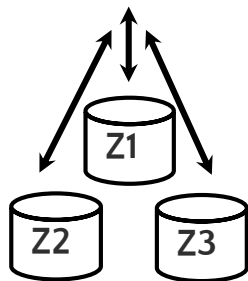
Determine Replication Strategies (1 of 2)

Single region



LRS

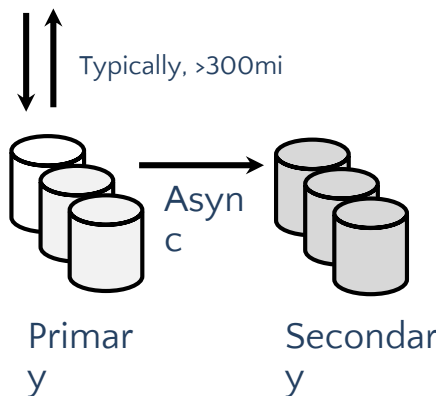
- Three replicas, one region
- Protects against disk, node, rack failures
- Write is acknowledged when all replicas are committed
- Superior to dual-parity RAID



ZRS

- Three replicas, three zones, one region
- Protects against disk, node, rack, and zone failures
- Synchronous writes to all three zones

Multiple regions

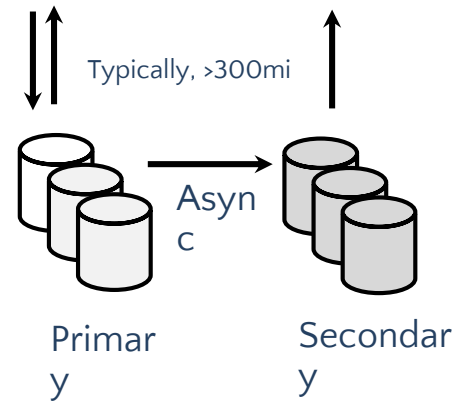


Primar
y

Secondar
y

GRS

- Six replicas, two regions (three per region)
- Protects against major regional disasters
- Asynchronous copy to secondary



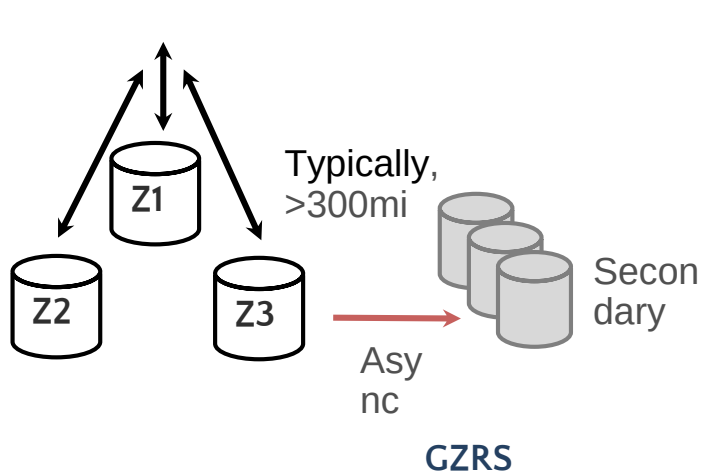
Primar
y

Secondar
y

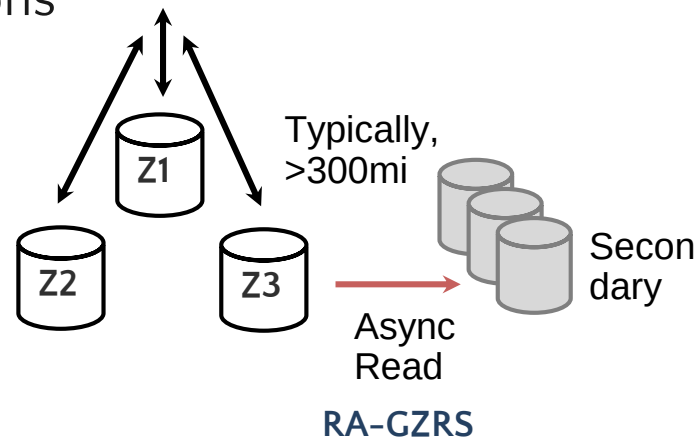
RA-GRS

- GRS + read access to secondary
- Separate secondary endpoint
- Recovery point objective (RPO) delay to secondary can be queried

Determine Replication Strategies (2 of 2)



- Six replicas, 3+1 zones, two regions
- Protects against disk, node, rack, zone, and region failures
- Synchronous writes to all three zones and asynchronous copy to secondary



- GZRS + read access to secondary
- Separate secondary endpoint
- RPO delay to secondary can be queried

Every object has a unique URL address – based on account name and storage type

Container service: `https://mystorageaccount.blob.core.windows.net`

Table service: `https://mystorageaccount.table.core.windows.net`

Queue service: `https://mystorageaccount.queue.core.windows.net`

File service: `https://mystorageaccount.file.core.windows.net`

If you prefer you can configure a custom domain name

Secure Storage Endpoints

storage987123 | Firewalls and virtual networks

Storage account

Search (Ctrl+ /)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Data transfer

Events

Storage Explorer (preview)

Save Discard Refresh

Allow access from

☐ All networks ☒ Selected networks

Configure network security for your storage accounts. [Learn more.](#)

Virtual networks

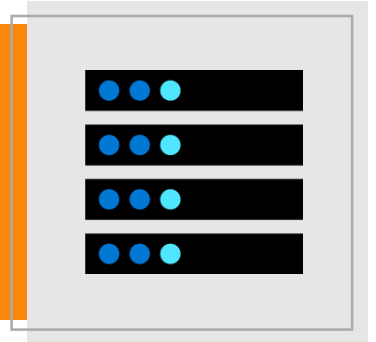
Secure your storage account with virtual networks. [+ Add existing virtual network](#) [+ Add new virtual network](#)

Virtual Network	Subnet	Address range	Endpoint Status	Resource Group
▼ vnet01	1			Demo
	subnet01	10.1.0.0/24	✓ Enabled	Demo

Firewalls and Virtual Networks restrict access to the Storage Account from specific Subnets on Virtual Networks or public IP's

Subnets and Virtual Networks must exist in the same Azure Region or Region Pair as the Storage Account

Lesson 02: Configure Blob storage



Implement Blob Storage

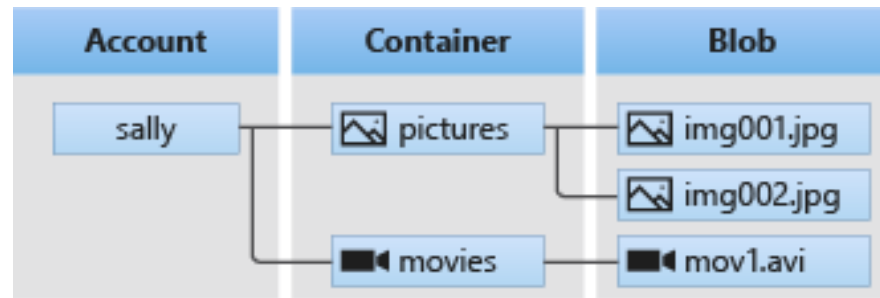
Stores unstructured data in the cloud

Can store any type of text or binary data

Also referred to as *object storage*

Common uses:

- Serving images or documents directly to a browser
- Storing files for distributed access
- Streaming video and audio
- Storing data for backup and restore, disaster recovery, archiving
- Storing data for analysis by an on-premises or Azure-hosted service



Create Blob Containers

All blobs must be in a container

Accounts have unlimited containers

Containers can have unlimited blobs

Private blobs – no anonymous access

Blob access – anonymous public read access for blobs only

Container access – anonymous public read and list access to the entire container, including the blobs

 Container  Change access level  Refresh |  Delete

New container

Name ^{*}



Public access level ⓘ



OK

Cancel

Public access level ⓘ



Private (no anonymous access)

Blob (anonymous read access for blobs only)

Container (anonymous read access for containers and blobs)

Create Blob Access Tiers

Hot tier – Optimized for frequent access of objects in the storage account

Cool tier – Optimized for storing large amounts of data that is infrequently accessed and stored for at least 30 days

Archive – Optimized for data that can tolerate several hours of retrieval latency and will remain in the Archive tier for at least 180 days

Access Tier

Optimize storage costs by placing your data in the appropriate access tier.

Hot (Inferred)	^
Hot (Inferred)	
Cool	
Archive	

You can switch between these access tiers at any time

Add Blob Lifecycle Management Rules

Transitioning of blobs to a cooler storage tier to optimize for performance and cost

Delete blobs at the end of their lifecycle

Apply rules to filtered paths in the Storage Account

Add a rule

✓ Details 2 Base blobs

Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).

+ Add if-then block

If

Base blobs were *

☒ Last modified

More than (days ago) *

Enter a value

↓

Then

Delete the blob

Move to cool storage

This is the most reliable option if cost is not a priority.

Move to archive storage

Archive storage does not fully delete the blob. However, it cannot be moved back to cool storage.

Delete the blob

This is the most efficient option if backing up a blob is not a priority.

Determine Blob Object Replication

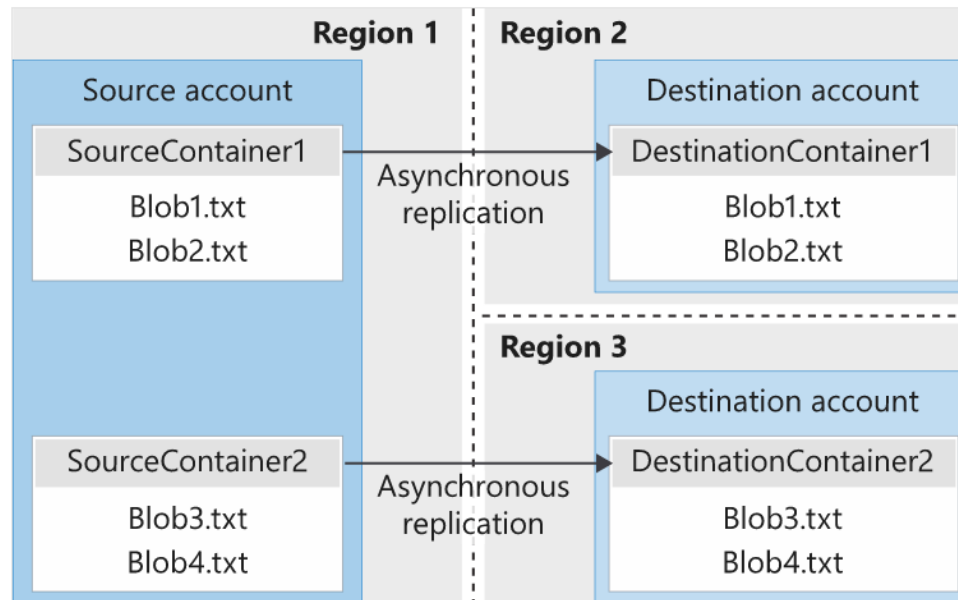
Asynchronous to any other Region

Minimizes latency for read requests

Increases efficiency for compute workloads

Optimizes data distribution

Optimizes costs



Upload Blobs

Authentication type – Azure AD user account or Account key

Block blobs (default) – useful for storing text or binary files

Page blobs – more efficient for frequent read/write operations

Append blobs – useful for logging scenarios

Upload blob

Files ⓘ

Select a file

☐ Overwrite if files already exist

^ Advanced

Authentication type ⓘ

Azure AD user account

Account key

Blob type ⓘ

Block blob

Block blob

Page blob

Append blob

☒ Upload .vhd files as page blobs (recommended)

Block size ⓘ

4 MB

Upload to folder

Encryption scope

☒ Use existing default container scope

☐ Choose an existing scope



You cannot change a blob type once it has been created

Determine Storage Pricing

Storage costs

Blob storage

Data access costs

Transaction costs

Geo-Replication data transfer costs

Outbound data transfer costs

Changing the storage tier

Block Blobs

Scalable object storage for documents, videos, pictures, and unstructured text or binary data. Choose from Hot, Cool, or Archive tiers.

Prices for locally redundant storage (LRS) Archive Block Blob start from:

\$0.002/GB per month

[See Pricing >](#)

Files

Fully managed file shares in the cloud, accessible via standard Server Message Block (SMB) protocol. Enables sharing files between applications using Windows APIs or REST API.

Prices for LRS File storage start from:

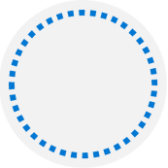
\$0.06/GB per month

[See Pricing >](#)

Lesson 03: Configure Storage Security



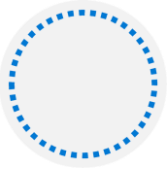
Review Storage Security Strategies



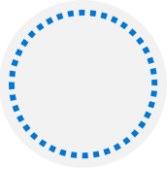
Storage Service Encryption



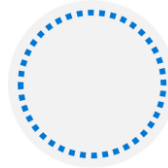
Authentication with Azure AD
and RBAC



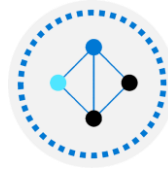
Client-side encryption, HTTPS,
and SMB 3.0 for data in transit



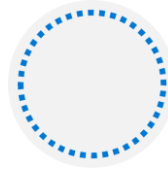
Azure disk encryption



Shared Access Signatures –
delegated access



Shared Key – encrypted
signature string



Anonymous access to
containers and blobs

Create Shared Access Signatures

Provides delegated access to resources

Grants access to clients without sharing your storage account keys

The account SAS delegates access to resources in one or more of the storage services

The service SAS delegates access to a resource in just one of the storage services

Signing method ⓘ

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1 ▼

Permissions * ⓘ

Read ▼

Start and expiry date/time ⓘ

Start

02/01/2021

(UTC-08:00) Coordinated Universal Time-08 ▼

Expiry

02/02/2021

(UTC-08:00) Coordinated Universal Time-08 ▼

Allowed IP addresses ⓘ

for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ

☒ HTTPS ☐ HTTP

Generate SAS token and URL

Identify URI and SAS Parameters

- A SAS is a signed URI that points to one or more storage resources
- Consists of a storage resource URI and the SAS token



`https://myaccount.blob.core.windows.net/?sp=r&st=2020-05-11T18:31:43Z&se=2020-05-12T02:31:43Z&spr=https&sv=2019-10-10&sr=b&sig=j0qABJZHfUvEBQ3yVn7kWICKl00sxCiK1rzEchfAz8U%3D`

Includes parameters for resource URI, storage services version, services, resource types, start time, expiry time, resource, permissions, IP range, protocol, signature

Determine Storage Service Encryption

Protects your data for security and compliance

Automatically encrypts and decrypts your data

Encrypted through 256-bit AES encryption

Is enabled for all new and existing storage accounts and cannot be disabled

Is transparent to users

Encryption

 Save  Discard

Storage service encryption protects your data at rest. Azure Storage encrypts your data as it's written in our datacenters, and automatically decrypts it for you as you access it.

By default, data in the storage account is encrypted using Microsoft Managed Keys. You may choose to bring your own key.

Please note that after enabling Storage Service Encryption, only new data will be encrypted, and any existing files in this storage account will retroactively get encrypted by a background encryption process.

[Learn More about Azure Storage Encryption](#) 

Encryption type

- ☒ Microsoft Managed Keys
☐ Customer Managed Keys



You can use your own key (next topic)

Create Customer Managed Keys

Use the Azure Key Vault to manage your encryption keys

Create your own encryption keys and store them in a key vault

Use Azure Key Vault's APIs to generate encryption keys

Custom keys give you more flexibility and control

Encryption type

- ☐ Microsoft Managed Keys
- ☒ Customer Managed Keys

i The storage account named 'storage987123' will be granted access to the selected key vault. Both soft delete and purge protection will be enabled on the key vault and cannot be disabled. [Learn more about customer managed keys](#) 

Encryption key

- ☐ Enter key URI
- ☒ Select from Key vault

Key vault and key *

Key vault: keyvault987123

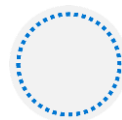
Key: storagekey

[Select a key vault and key](#)

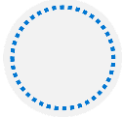
Apply Storage Security Best Practices



Always use HTTPS to create or distribute an SAS



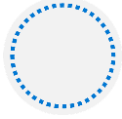
Be specific with the resource to be accessed



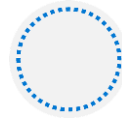
Reference stored access policies where possible



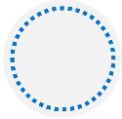
Understand that your account will be billed for any usage



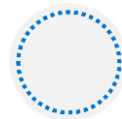
Use near-term expiration times on an ad hoc SAS



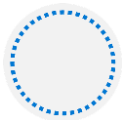
Validate data written using SAS



Use Storage Analytics to monitor your application



Don't assume SAS is always the correct choice



Be careful with SAS start time

Lesson 04: Configure Azure Files and File Sync



Compare Files to Blobs

Feature	Description	When to use
Azure Files	SMB interface, client libraries, and a REST interface that allows access from anywhere to stored files	• Lift and shift an application to the cloud • Store shared data across multiple virtual machines • Store development and debugging tools that need to be accessed from many virtual machines
Azure Blobs	Client libraries and a REST interface that allows unstructured data (flat namespace) to be stored and accessed at a massive scale in block blobs	• Support streaming and random-access scenarios • Access application data from anywhere

Manage File Shares

File share quotas

Windows – ensure port 445 is open

Linux – mount the drive

MacOS – mount the drive

Secure transfer required – SMB 3.0 encryption

Windows Linux macOS

Drive letter

Z

To connect to this Azure file share from Windows, run these PowerShell commands from a normal (not elevated) PowerShell terminal:

```
$connectTestResult = Test-NetConnection -  
ComputerName storage987123.file.core.windows.net -  
Port 445  
if ($connectTestResult.TcpTestSucceeded) {  
    # Save the password so the drive will persist on reboot  
    cmd.exe /C "cmdkey  
    /add:"storage987123.file.core.windows.net"
```

This script will check to see if this storage account is accessible via TCP port 445, which is the port SMB uses. If port 445 is available, your Azure file share will be persistently mounted. Your organization or internet service provider (ISP) may block port 445, however you may use Azure [Point-to-Site \(P2S\) VPN](#), Azure [Site-to-Site \(S2S\) VPN](#), or [ExpressRoute](#) to tunnel SMB traffic to your Azure file share over a different port.

Create File Share Snapshots

 Add snapshot  Refresh  Delete		
Name	Date created	Initiator
☐ 2020-03-12T00:58:38.0000000Z	3/11/2020, 8:58:38 PM	-

Incremental snapshot that captures the share state at a point in time

Is read-only copy of your data

Snapshot at the file share level, and restore at the file level

- Protection against application error and data corruption
- Protection against accidental deletions or unintended changes
- General backup purposes

Identify File Sync Components

The **Storage Sync Service** is the top-level resource

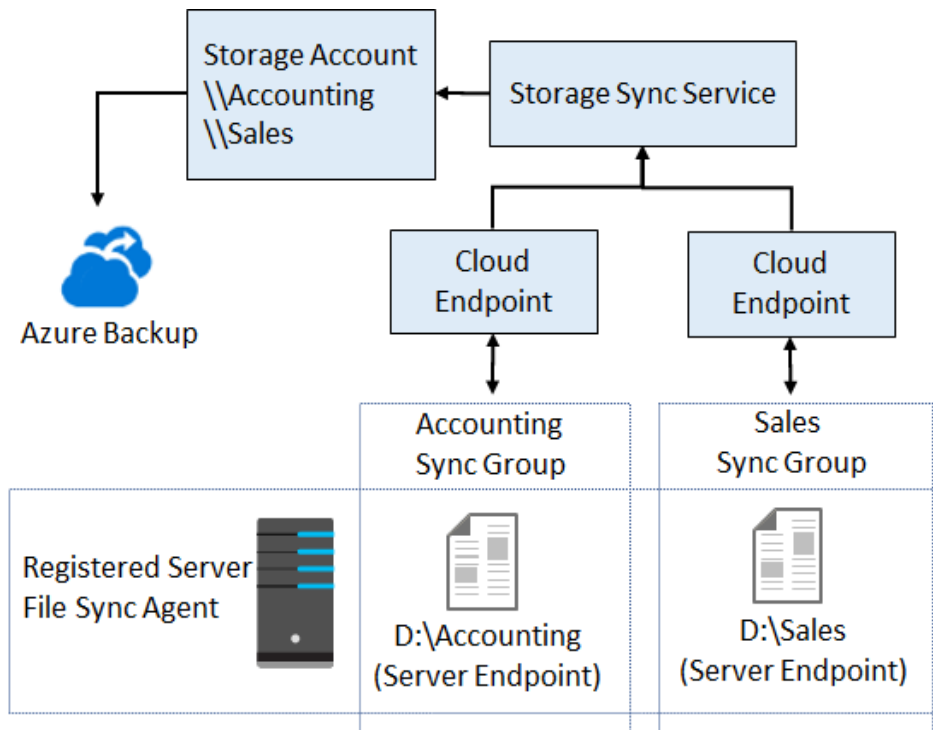
The **registered server** object represents a trust relationship between your server (or cluster) and the Storage Sync Service

The **Azure File Sync agent** is a downloadable package that enables Windows Server to be synced with an Azure file share

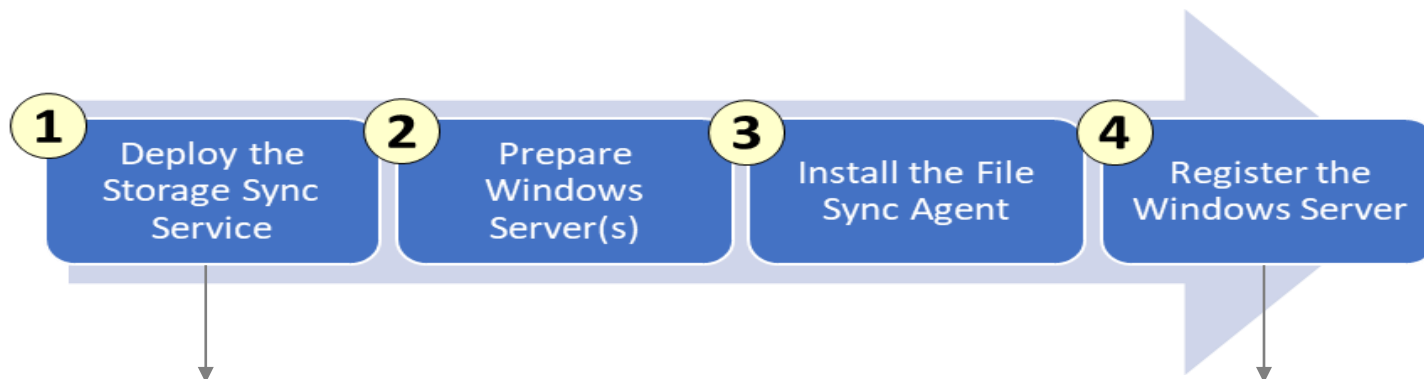
A **server endpoint** represents a specific location on a registered server, such as a folder

A **cloud endpoint** is an Azure file share

A **sync group** defines which files are kept in sync



Setup File Sync



Home > Deploy Storage Sync

Deploy Storage Sync □ ×

* Name
StorageSync1 ✓

* Subscription
Visual Studio Enterprise ▾

* Resource group
ASH ▾

Create new

* Location
South Central US ▾

Create Automation options

Microsoft Azure File Sync - Server Registration

Choose a Storage Sync Service

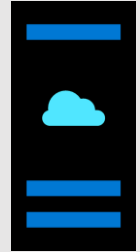
Azure Subscription
▾

Resource Group
▾

Storage Sync Service
▾

Register

Lesson 05: Configure Storage with Tools



Use Storage Explorer

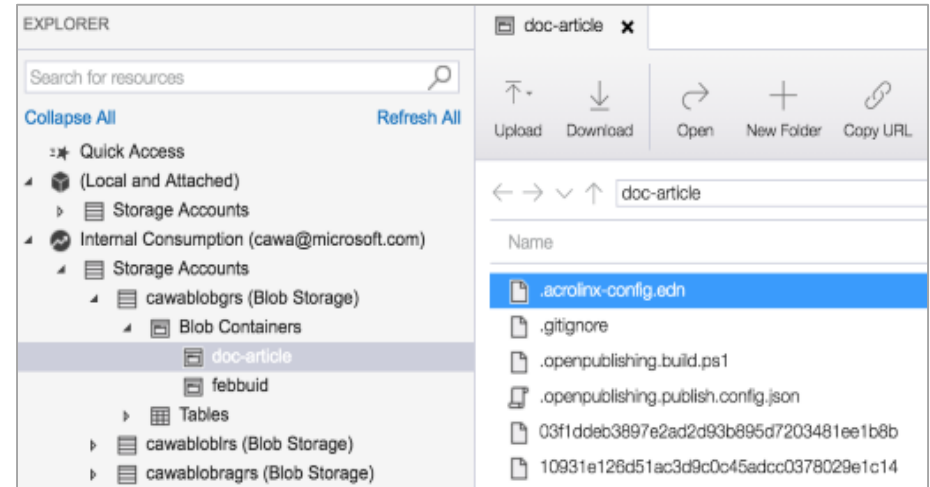
Access multiple accounts and subscriptions

Create, delete, view, edit storage resources

View and edit Blob, Queue, Table, File, Cosmos DB storage and Data Lake Storage

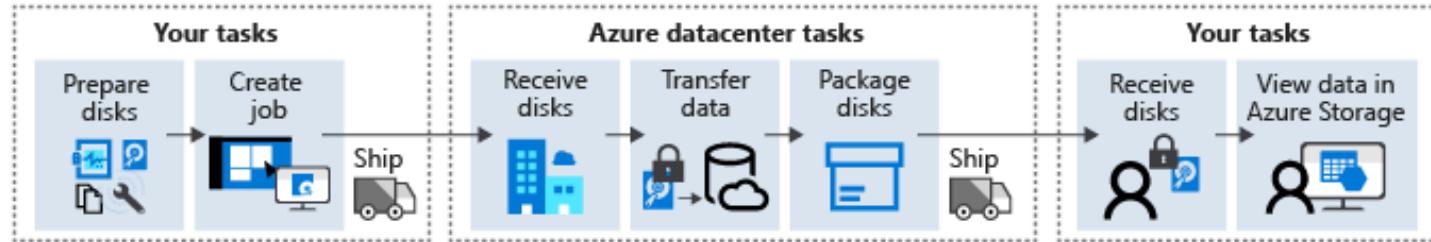
Obtain shared access signature (SAS) keys

Available for Windows, Mac, and Linux

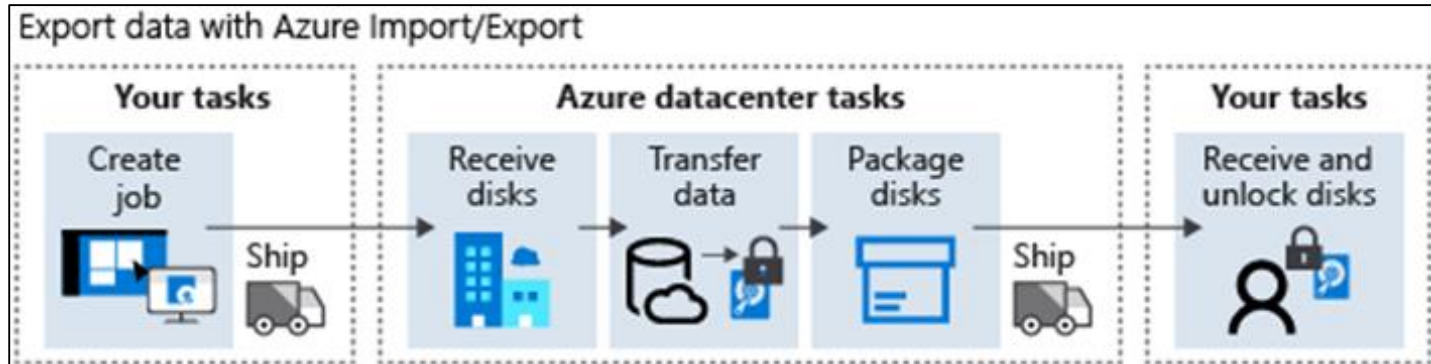


Use the Import and Export Service

Import jobs move large amounts of data to Azure blob storage or files



Export jobs move large amounts of data from Azure blob storage (not files)



```
azcopy copy [source] [destination] [flags]
```

Command line utility

Designed for copying data to and from
Azure Blob, File, and Table storage

Available on Windows, Linux, and MacOS

Authentication options include Active
Directory or SAS token

Thank you

